Robustness of the I-Bug Algorithm Against Adversarial Signal Spoofing in Superposition Environments

Tsuriel vizel Avraham Guez

Supervisor: Dr. A i Zhitnikov

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Abstract

Sensor-based bug algorithms provide robust navigation guarantees using minimalist sensing. However, algorithms relying purely signal intensity, such as the I-Bug algorithm, are highly vulnerable to adversarial sp g. In realistic physical environments, signals from multiple sources exhibit superposition rather than simple displace . When a malicious tower broadcasts intermittent signals, these spoofed waves superimpose on the true target signal, creating complex temporal and spatial distortions that misdirect the robot. This paper introduces an anomaly-detecting extension to the I-Bug algorithm that preserves target convergence in the presence of an active adversary. By differentiating between spatial intensity gradients and temporal signal discontinuities, our system detects additive spoofing attacks. We propose and evaluate three behavioral strategies: a Naive approach, a Cautious (pausing) strategy, and an Inertial (semi-cautious) heuristic. We mathematically prove that only the Cautious strategy rigorously guarantees arrival at the target in a finite number of steps while preserving the original spatial path bounds. To empirically validate our theoretical findings, we present a comprehensive simulation modeling physical wave superposition across randomized environments. Our statistical results demonstrate that while the Naive approach fails entirely and the Inertial heuristic suffers from diminished success rates, the Cautious algorithm consistently achieves the target, highlighting a necessary trade-off between execution time and guaranteed convergence.

1 Introduction

Bug algorithms are a family of sensor-based planning algorithms that allow robots to navigate toward a goal in unknown environments using minimal sensory input. The I-Bug algorithm represents a minimalist extreme, requiring only an intensity sensor that measures the signal strength emanating from a target tower, without access to position coordinates or odometry.

While mathematically robust under ideal conditions, I-Bug operates on the assumption that the target tower is the sole source of the signal. In real-world hostile environments, an adversary may deploy an "enemy tower" to spoof the signal. Furthermore, physical signals (such as RF or acoustic waves) follow the principle of superposition.

Rather than simply overriding the target signal, the adversary's transmission adds to the total perceived intensity, creating a combined signal field that tricks the robot into infinite loops or directs it into obstacles. This paper addresses this vulnerability by introducing an anomaly-detection filter and exploring two distinct behavioral counter-strategies: the *Cautious Strategy* and the *Inertial Strategy*. We prove that the Cautious framework maintains the strict convergence guarantees of the original algorithm and provide an empirical analysis of all strategies using a custom simulation environment modeled on wave superpositio

2 Problem Formulation

Consider a point robot moving in a 2D environment $\mathbb{R}^2$ containing unknown, piecewise-analytic obstacles. A target tower located at p_t emits a signal represented by the intensity function $m(p-p_t)$, which increases monotonically as the distance to the tower decreases. The robot operates using three motion primitives: u_{fwd} (move forward), u_{ori} (orient toward the strongest signal), and u_{fol} (follow an obstacle boundary).

We introduce an adversarial enemy tower located at an unknown position. The enemy tower broadcasts a spoofing signal intermittently. We model the signal propagation using the physical principle of wave superposition.

Assumption 1. *As established in the original I-Bug framework, $m(p-p_t)$ is assumed to be a locally Lipschitz, piecewise-analytic function. Let the total received signal intensity at position p and time t be defined as $S(p,t) = m(p-p_t)+e(t)$, where $e(t) \geq 0$ represents the enemy tower's intermittent transmission. Unlike simplistic models that assume the receiver only detects the strongest independent signal, this additive formulation accurately reflects the physical superposition of signal intensities.*

Assumption 2. *Because the robot's spatial velocity is bounded and m is locally Lipschitz, the temporal derivative of $m(p-p_t)$ due to normal robot motion is strictly bounded. We assume $e(t)$ acts as a step function when transitioning states, creating an instantaneous discontinuity ($\frac{\partial S}{\partial t} \to \infty$). This allows the robot to distinctly differentiate between a spatial gradient change (normal movement) and a temporal jump (the additive burst of an enemy transmission).*

3 The Proposed Anti-Adversary Algorithm

To counter the spoofing attack, we extend the I-Bug algorithm with an anomaly detection filter that calculates the maximum physically possible intensity increase for a single movement step. If the measured intensity change exceeds this physical limit, the system flags a positive anomaly (the adversary turned ON).

Upon detection, the robot employs one of the following counter-strategies:

3.1 The Naive Strategy

The robot applies no filter and continuously follows the highest perceived superimposed intensity, updating its internal maximum intensity memory blindly. This serves as our control baseli

3.2 The Cautious Strategy

Upon detecting a positive anomaly, the robot immediately halts all spatial movement ($\frac{dp}{dt} = 0$) and suspends updates to its internal state memory (the departure intensity i_L and hit intensity i_H). It waits in this paused state until a negative anomaly (a sharp signal drop corresponding to $-e(t)$) is detected and confirmed over consecutive steps. Once confirmed, it resumes its exact previous motion primitive based solely on the isolated $m(p - p_t)$ signa

3.3 The Inertial (Semi-Cautious) Strategy

Upon detecting a positive anomaly, the robot freezes its angular heading but continues to execute its active movement primitive (e.g., moving forward blindly or hugging a boundary). During boundary following, it skips the intensity-based leave-obstacle check, as the superimposed readings are compromised. It resumes normal sensor usage only when the signal drops

4 Mathematical Guarantees

We focus our rigorous mathematical analysis on the *Cautious Strategy*, which is the only heuristic capable of strictly preserving the original algorithm's spatial properties under superposition.

Theorem 1. *Under Assumptions 1 and 2, a robot utilizing the Cautious Strategy will reach the target tower in a finite number of steps if a valid path exists. The upper bound on the total distance traveled is identical to that of the original I-Bug algorithm, and the upper bound on the total time increases by the total duration during which the enemy tower was transmitting.*

Proof. While the robot is in its normal navigation state, it operates strictly according to the motion primitives and rules of the original I-Bug algorithm.

When the enemy tower begins transmitting, the robot detects the temporal discontinuity caused by the additive signal $e(t)$ and halts its movement. Because the spatial velocity is strictly zero in this state ($\frac{dp}{dt} = 0$), the robot's spatial position p and orientation θ remain completely unchanged for the entire duration of the enemy's transmission.

Crucially, because the internal state variables i_L and i_H are locked, the logical flow of the algorithm is perfectly preserved. If the robot was executing a boundary-following primitive (u_{fol}) when the enemy transmission began, it resumes the exact same u_{fol} primitive upon cessation of the signal, ensuring that the termination conditions for navigating obstacle boundaries are unaffected.

Once the enemy tower ceases its transmission, the additive component $e(t)$ drops to zero, and the measured signal intensity returns exactly to the original value $m(p-p_t)$. The robot then resumes its movement from the exact same spatial coordinates and internal state it maintained prior to the disturbance.

Consequently, the overall spatial trajectory traversed by the robot is perfectly identical to the geometric path it would have taken had the enemy tower not existed.

In the original paper, Proposition 2 establishes that the I-Bug algorithm guarantees convergence to the target tower in a finite number of steps. Because the spatial path in the

Cautious Strategy is geometrically identical to the original I-Bug path, this convergence guarantee is strictly preserved.

Conversely, the *Inertial Strategy* lacks this guarantee. By moving blindly and skipping boundary-leave checks during spoofed periods, the robot's spatial trajectory fundamentally deviates from the original path. This deviation can result in missing optimal departure points or becoming permanently trapped in non-convex obstacle geometries, voiding the theoretical guarantees of Proposition □

5 Simulation Setup and Methodology

To empirically validate the proposed algorithms, a custom 2D kinematic simulation was developed using the Julia programming language. The simulation modeled the signal intensity mapping $m(p)$ alongside rigid collision boundaries and incorporated physical wave superposition, where the total perceived intensity is the arithmetic sum of the target and adversary signals ($S(p,t) = m(p - p_t) + e(t)$).

The three distinct robot strategies (Naive, Cautious, and Inertial) were executed simultaneously, originating from the same starting coordinates. The testing methodology involved running the simulation across multiple randomized obstacle layouts. The adversary tower operated on a standardized duty cycle, emitting a signal strength that superimposed on the true target's signal.

6 Statistical Results

To evaluate the efficacy and convergence rates of the proposed strategies in a superposition environment, we executed 1,000 randomized simulation runs.



Figure 1: Success rates of the Naive, Cautious, and Inertial strategies across 1,000 randomized superposition environments.

As shown in Figure 1, the Naive robot suffered a near 0% success rate, as the combined signal gradient repeatedly drew it toward the adversary tower. The Cautious robot achieved a 100% success rate, corroborating our mathematical proof of convergence. The Inertial strategy yielded mixed results; while it occasionally reached the target faster by moving blindly through open spaces, its lack of spatial guarantees caused it to fail in complex obstacle configurations.

Strategy	Success Rate	Average Distance (Successful Runs)	Average Steps (Time)
Naive	[INSERT %]	N/A (Failed)	N/A (Failed)
Cautious	100%	[INSERT DATA]	[INSERT DATA]
Inertial	[INSERT %]	[INSERT DATA]	[INSERT DATA]

Table 1: Comparison of navigation efficiency between the strategies under additive signal spoofing.

(Discuss the data from Table 1 here. Emphasize the trade-off: The Cautious strategy guarantees success but requires more time (steps) due to pausing. The Inertial strategy might save time in simple environments but risks catastrophic failure and path divergence in complex environments).

7 Limitations and Future Work

While the Cautious algorithm provides strong theoretical guarantees against intermittent spoofing, the current framework possesses limitations that warrant future research:

- **Sensor Noise:** The current anomaly detection relies on identifying an absolute mathematical discontinuity ($\frac{\partial S}{\partial t} \to \infty$). In physical hardware, high Gaussian noise or signal multipath fading could trigger false positives. Future iterations should incorporate probabilistic filtering to smooth sensor readings before applying anomaly thresholds.
- **Ramping Adversary Attack:** The algorithm assumes the adversary tower operates as a step function. If an intelligent adversary gradually ramps up its signal strength to mimic the robot's maximum physical movement speed, the anomaly detection filter would fail to trigger, allowing the superimposed signal to seamlessly hijack the robot's trajectory.
- **Static Adversary Assumption:** The mathematical proof assumes the enemy tower is stationary. The dynamics of a mobile adversary actively pursuing the robot require a different theoretical approach.

8 Conclusion

This paper addressed a critical vulnerability in intensity-based bug algorithms: susceptibility to adversarial signal spoofing in physical environments governed by wave superposition. We demonstrated that a minimalist sensor suite is capable of rejecting additive adversarial interference by extracting temporal cues from spatial signal data. Through the introduction of the Cautious anti-adversary strategy, we proved that target convergence and path bounds can be strictly maintained by suspending spatial movement during spoofed periods. Our simulation results validated these theoretical claims, showing a 100% success rate for the Cautious approach, while highlighting the inherent failure risks associated with naive superimposed signal tracking and blind inertial heuristics.

References

[1] Kamilah Taylor and Steven M. LaValle, "I-Bug: An Intensity-Based Bug Algorithm," *IEEE International Conference on Robotics and Automation*, 2011.